Lecture 40: Deep Learning (Part-5)

CIFAR-10 CNN, NLP Foundations, Word Embeddings, RNN & LSTM

Welcome to the comprehensive, publication-grade documentation for **Lecture 40: Deep Learning (Part-5)**. This curriculum covers end-to-end computer vision implementations in PyTorch, historical architectural evolutions in CNNs, classical text representations, continuous distributed word embeddings, recurrent neural architectures, backpropagation through time dynamics, and gated recurrent memory units.

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1. Master Curriculum Architecture

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lecture_40/
├── Lecture_Code/
│   ├── CNN_for_CIFAR10.ipynb             <- End-to-end executable PyTorch notebook
│   └── Resources/
│       ├── 5 Popular CNN Architectures.pdf <- Landmark CNN reference
│       ├── Word2Vec Reading Material.png  <- Vector semantics reference
│       └── LSTM Reference Pages           <- Hochreiter & Schmidhuber reference
│
├── Notes_40.01/  <- CIFAR-10 Dataset & PyTorch Preprocessing Pipeline
├── Notes_40.02/  <- Building the 3-Block Deep CNN Architecture in PyTorch
├── Notes_40.03/  <- CNN Training, Optimization & Test Evaluation on CIFAR-10
├── Notes_40.04/  <- Landmark CNN Architectures: From LeNet-5 to ResNet
├── Notes_40.05/  <- Foundations of Natural Language Processing (NLP) & Text Preprocessing
├── Notes_40.06/  <- Text Representation: One-Hot Encoding, Bag-of-Words (BoW) & N-Grams
├── Notes_40.07/  <- Advanced Text Representation: TF-IDF & Distributed Word Embeddings (Word2Ve
├── Notes_40.08/  <- Recurrent Neural Networks (RNN): The Recurrent Loop & Time Unrolling
├── Notes_40.09/  <- Backpropagation Through Time (BPTT), Vanishing & Exploding Gradients
├── Notes_40.10/  <- Long Short-Term Memory (LSTM) Networks: Cell State & Gating Architecture
│
└── Practice_Problems/
    ├── README.md                         <- 15 High-Yield Technical Problems (English)
    ├── README_HINGLISH.md                <- 15 High-Yield Technical Problems (Hinglish)
    ├── Solutions.md                      <- Full Mathematical & Code Solutions (English)
    └── Solutions_HINGLISH.md             <- Full Mathematical & Code Solutions (Hinglish)
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2. Overview of Modules

Submodule	Core Theoretical & Practical Concepts	Key Formulas / Architectures	Visual Artifact
Notes_40.01	CIFAR-10 Dataset (60K $32 \times 32 \times 3$ images), DataLoaders ($B = 64$, 782 batches),	$x_{\text{norm}} = \frac{x-0.5}{0.5} \in [-1, 1]$	`cifar10_dataset_preprocessing_pipeline.png`

Submodule	Core Theoretical & Practical Concepts	Key Formulas / Architectures	Visual Artifact
	Normalization pipeline		
Notes_40.02	3-Block Deep CNN in PyTorch, spatial halving ($32 \rightarrow 16 \rightarrow 8 \rightarrow 4$), flattening ($128 \times 4 \times 4 = 2,048$)	$620,362$ learnable parameters	`deep_cnn_cifar10_architecture.png`
Notes_40.03	Training loop, Adam optimizer, `CrossEntropyLoss`, test evaluation, `model.eval()`, `torch.no_grad()`	Test Accuracy $\approx 72\% - 75\%$	`cnn_training_and_evaluation_workflow.png`
Notes_40.04	Landmark CNNs: LeNet-5, AlexNet ReLU, VGG 3×3 factorization, GoogLeNet Inception, ResNet identity shortcuts	$\mathbf{y} = \mathcal{F}(\mathbf{x}) + \mathbf{x} \implies \frac{\partial \mathcal{E}}{\partial \mathbf{x}} = \frac{\partial \mathcal{E}}{\partial \mathbf{y}} \left(\frac{\partial \mathcal{F}}{\partial \mathbf{x}} + \mathbf{I} \right)$	`landmark_cnn_architectures_evolution.png`
Notes_40.05	NLP Foundations, Tokenization, Stopwords, Stemming (Porter) vs Lemmatization (WordNet)	Syntax vs Semantics	`nlp_text_preprocessing_pipeline.png`
Notes_40.06	One-Hot Encoding, Bag-of-Words (BoW), Document-Term Matrix (DTM), N-Grams context modeling	Orthogonality ($\mathbf{e}_i \cdot \mathbf{e}_j = 0$), Negation handling	`text_representation_onehot_bow_ngrams.png`
Notes_40.07	TF-IDF statistical weighting, Distributed Word Embeddings, Word2Vec CBOW vs Skip-Gram	$\text{TF} \times \text{IDF}$, $\vec{\text{King}} - \vec{\text{Man}} + \vec{\text{Woman}} \approx \vec{\text{Queen}}$	`tfidf_and_word_embeddings.png`
Notes_40.08	Recurrent Neural Networks, time unrolling, hidden	$\mathbf{h}_t = \tanh(\mathbf{W}_{xh}\mathbf{x}_t +$	`rnn_recurrent_neuron_unrolling.png`

Submodule	Core Theoretical & Practical Concepts	Key Formulas / Architectures	Visual Artifact
	state recurrence, 5 sequence architectures	$\mathbf{W}_{hh}\mathbf{h}_{t-1} + \mathbf{b}_h)$	
Notes_40.09	Backpropagation Through Time (BPTT), Vanishing Gradients ($\|\lambda\| < 1$), Exploding Gradients ($\|\lambda\| > 1$), Gradient Clipping	$\mathbf{g} \leftarrow \theta \frac{\mathbf{g}}{\|\mathbf{g}\|_2}$	`bptt_vanishing_exploding_gradients.png`
Notes_40.10	LSTM Cell Architecture, Constant Error Carousel (CEC), Forget Gate, Input Gate, Output Gate	$\mathbf{C}_t = \mathbf{f}_t \odot \mathbf{C}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{C}}_t$	`lstm_cell_architecture_gating.png`

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3. Educational Handwritten Visual System

Every single concept across this curriculum is accompanied by a **notebook-friendly, hand-drawn educational diagram** created in standard blue/black pen notebook styling with clean annotations, mathematical labels, and step-by-step unrolled flows. All instructor webcam and video keyframe images have been strictly omitted.